data / reporting
uses training rows only
held-out test (opened once)
1 Raw data
UCI #501 · Aotizhongxin station
35,064 hourly rows × 18 columns
2 Parse & validate
datetime from year/month/day/hour
sort · regular hourly grid (0 gaps)
3 Target + time split
target = PM2.5(t+1); last row dropped
80 % train | 20 % test (locked)
no shuffling
4 Causal features
current PM2.5 · lags · shifted rolling
calendar · differences · interactions
5 EDA + hypotheses
training rows only; H1–H4 and the
decision rules pre-registered
6 Feature sets
A raw · B top-k correlated ·
C engineered (leakage-tested)
7 TimeSeriesSplit CV
5 expanding folds; imputer, scaler
and top-k fitted inside each
training fold
8 Model selection
OLS / Ridge / Lasso / Elastic Net
CV-MAE; 1-SE tie → rule from
the data properties
9 Ablation & inference
A/B/C + group removal;
HAC OLS · bootstrap CI · VIF
10 Freeze design
feature lists, hyper-parameters,
folds, seed → SHA-256
11 Open test set once
predict with frozen models;
MAE · RMSE · R² · DM tests
12 Slices, errors, residuals
slice analysis, 3 largest errors,
residual diagnostics